Unsupervised and Unpaired Fusion-Based Hyperspectral Image Super-Resolution Based on Reflectance Migration

Wei Liu, Haixia Bi, *Member, IEEE*, Xuehu Zhu, Chen Xu, and Bin Pan, *Member, IEEE*

***Abstract*— Fusion-based hyperspectral image (HSI) super-resolution has garnered growing attention in recent years due to its strong capability in reconstructing high spatial resolution (HR) HSIs. However, most existing methods either rely on a supervised training strategy or require pairwise images, which are difficult to acquire. To address these issues, we propose an unsupervised and unpaired fusion-based super-resolution network, called U^{2}SRnet, which achieves super-resolution without the aid of any supervision nor pairwise images. Instead, an arbitrary low spatial resolution (LR) HSI is used as a fixed hyperspectral input to provide reflectance information for the spectral reconstruction of RGB images. U^{2}SRnet consists of two modules, which are the spectral correction module (SCM) and band generation module (BGM), respectively. The former aims to enforce material reflectance consistency between unpaired HS–RGB images, which is implemented via a joint perception attention module (JPAM). Considering the spatial inconsistency of material positions between HS and RGB images, we propose an elaborately designed material-matching mechanism in BGM to obtain HS–RGB block pairs with high material-semantic similarity. Furthermore, a differential migration mechanism guided by physical reflectance priors, universally applicable to unpaired images, is introduced to migrate the material reflectance properties between these block pairs. Extensive experiments were performed on Cave, Harvard, and TG1HRSSC datasets. The experimental results validated the effectiveness and superiority of the proposed method.**

***Index Terms*— Fusion-based super-resolution, hyperspectral image (HSI), unpaired, unsupervised deep learning (DL).**

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Wei Liu and Haixia Bi are with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: liuwei_xjtu@stu.xjtu.edu.cn; haixia.bi@xjtu.edu.cn).

Xuehu Zhu is with the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: zhuxuehu@xjtu.edu.cn).

Chen Xu is with the Department of Mathematics and Fundamental Research, Peng Cheng Laboratory, Shenzhen 518055, China, and also with the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: xuch01@pcl.ac.cn).

Bin Pan is with KLMDASR, LEBPS, LPMC, School of Statistics and Data Science, Nankai University, Tianjin 300071, China (e-mail: panbin@nankai.edu.cn).

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I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spectral information via imaging across a multitude of near-continuous spectral bands, which allows for detailed analysis of different materials and objects. HSIs have been extensively used in various applications, such as target detection, disaster monitoring, and military reconnaissance [1]. However, limited by imaging techniques, it is difficult to obtain high spatial resolution (HR) HSIs [2], [3], [4]. In this scenario, a promising research direction has emerged, i.e., hyperspectral super-resolution (HSSR).

Considering the input data employed, HSSR can be classified into three branches: spatial super-resolution of HSIs, spectral super-resolution of RGB images, and fusion-based super-resolution. Spatial super-resolution focuses on recovering spatial texture details of the low spatial resolution (LR) HSIs [5], [6], [7], while spectral super-resolution aims to generate HR HSIs by reconstructing spectral channels from RGB images [8], [9], [10], [11]. Since both types of approaches are performed on single-modality images, they are collectively referred to as single-image super-resolution (SISR). Despite the emergence of massive amounts of work on SISR in recent years, constrained by insufficient information from a single modality, these methods typically require supervision from the ground-truth HR HSIs.

Utilizing both HSIs and RGB images as inputs, fusion-based super-resolution aims to generate high-quality images by integrating complementary information from multimodality images, which indicates that it can work in an unsupervised manner, without relying on ground-truth HR HSIs. Although significant progress has been witnessed, most fusion-based methods, including convolutional neural network (CNN)-based approaches [12], [13], [14], [15], [16] and transformer-based architectures such as FusFormer [17], MSST-Net [18], and uHNTC [19], rely on an essential prerequisite: paired HS–RGB images. However, acquiring such paired data is both time-consuming and expensive, limiting practical applications of these methods. Therefore, developing an unsupervised super-resolution approach capable of handling unpaired HS–RGB images is essential, which is precisely the focus of this work.

Previous studies [16], [20], [21] have revealed that the same materials tend to exhibit similar spectral reflectance properties

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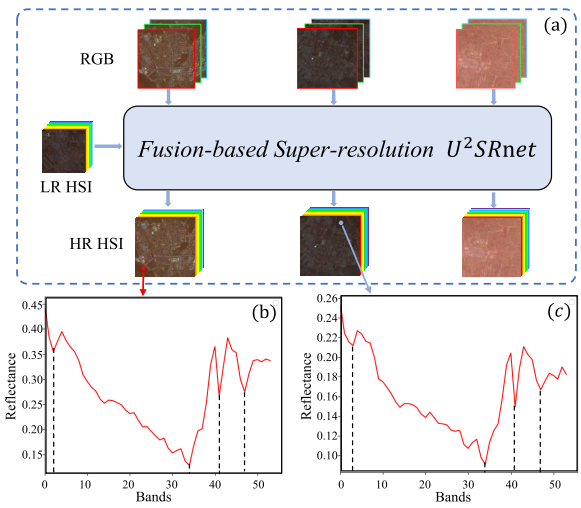


Fig. 1. (a) Illustration of network inputs and outputs. (b) and (c) Reflection curve and absorption peaks of grass materials selected from two different images. The wavelength bands corresponding to the absorption peaks are fixed, whereas the values at these peaks and the trends of the curves at nonabsorption peaks vary.

in the same imaging environment, even across different scenes. This principle signifies that reflectance properties of materials can be migrated from unpaired LR HSIs to RGB images to guide the super-resolution, as long as they contain the same materials. Building on this prior knowledge, we are motivated to explore the feasibility of reconstructing HR HSIs from RGB images by fusing unpaired HS–RGB images. Specifically, an arbitrary LR HSI is utilized as a fixed HSI input to provide material reflectance information applicable to various RGB images, as illustrated in Fig. 1(a).

Reflectance properties can be described by absorption-band parameters, including position, intensity, and bandwidth [21]. Notably, the positions of absorption peaks, determined by the chemical structure of materials, remain invariant regardless of the imaging environments or scenes. However, the intensities at absorption peaks and the curve trends at nonabsorption peaks may vary due to changes in external imaging environments, such as the angle of light incidence and the position of sensors, as illustrated in Fig. 1(b) and (c). Under these circumstances, crudely migrating reflectance properties would introduce noise interference [22]. Therefore, the first challenge we need to address for unpaired super-resolution is *how to achieve material reflectance consistency between LR HSI and RGB images*.

Due to the spatial inconsistency of material positions in unpaired images, traditional spatial position-based cropping and block matching methods, as employed in previous studies [12], [13], [14], [15], [16], are definitely infeasible in this scenario. To this end, for each spatial block in the RGB image, it is necessary to first identify the block most likely to contain identical materials in the LR HSI. Given the material-matched block pairs, the consequent task is to design a generalizable reflectance property migration mechanism, which can be applied to various unpaired images. In summary, the second challenge lies in *how to effectively match and migrate reflectance properties between unpaired HS–RGB images*.

In this article, we propose an unsupervised and unpaired fusion-based super-resolution network (U^{2}SRnet) for HSSR, which targets practical remote sensing scenarios where HR HSIs are scarce, and only unpaired RGB and LR HS images are available. The RGB images provide spatial information and serve as targets for spectral enhancement. In contrast, a single LR HSI is used as the fixed hyperspectral input, offering spectral reflectance priors to guide the reconstruction. U^{2}SRnet comprises two components, spectral correction module (SCM) and band generation module (BGM), each addressing one of the aforementioned challenges. Environmental factors influencing spectral reflectance are often simplified as random noise in existing methods, which limits the effectiveness of fusion-based super-resolution, particularly under varying imaging environments. In the SCM, we devise a joint perception attention module (JPAM), where the attention weights encode the spectral divergence caused by imaging environment differences between unpaired HS–RGB images. By integrating the inferred attention map into the high-dimensional spectral features, we correct the spectral features in the LR HSI, enforcing reflectance consistency between two modalities. After correcting environmental influences, spectral reflectance priors, which have been neglected in previous unpaired fusion-based methods [7], [20], [23], are essential for reliably migrating material reflectance properties. To utilize these priors, the BGM is introduced and designed to operate within a hybrid feature-data domain. The inputs are first embedded into a shared feature domain, and a new nonnegative Pearson correlation coefficient (CC) network is then devised to identify HS–RGB block pairs with high material-semantic similarity from unpaired images. Furthermore, a differential migration mechanism in the data domain is designed to extract and migrate the reflectance intensity variation information of materials between block pairs. Specifically, residual maps of the LR HSI block, as a differential representation of reflectance properties, are integrated with the RGB block (as reference bands) through migration modules and shared weights to reconstruct spectral bands of RGB block. In addition, we introduce a pooling spectral angular loss to constrain reconstructed HR HSIs, effectively preventing anomalies in reflectance peak intensities. To the authors' knowledge, the whole super-resolution framework, combined with both the SCM and BGM are new designs, which have never been presented in existing works.

Our main novelties and contributions are summarized as follows.

1) We propose U^{2}SRnet, which generates HR HSIs via migrating material reflectance properties between unpaired HS–RGB images, overcoming the difficulty of obtaining pairwise images.
2) We introduce a JPAM that converts spectral divergence features into environment-aware attention weights, enabling adaptive correction of LR HSI for consistent material reflectance.
3) We propose a hybrid feature-data domain BGM that integrates dynamic learning with physical interpretability through material-semantic matching in a shared feature space and data-domain differential migration, enhancing fusion quality and generalization.

The remainder of this article is organized as follows. Section II introduces the related works of HSSR from the perspective of inputs. We, then, elaborate on the proposed

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method in Section III. The experimental results and analysis are presented in Section IV. We finally conclude this article and discuss the future works in Section V.

II. Related Works

A. Spatial Super-Resolution of HSI

Spatial super-resolution aims to generate HR HSIs by reconstructing the spatial details of LR HSIs. Early works explicitly formulate spatial super-resolution task as a constrained optimization problem, employing priors to regularize the solution space. For instance, Huang et al. [24] used a low-rank and group-sparse model to achieve spatial super-resolution. Zhang et al. [25] proposed an HSI super-resolution algorithm based on the maximum a posteriori (MAP). Most existing deep learning (DL) super-resolution algorithms recover the missing high-frequency details using CNNs [5], [26], [27] or attention mechanisms [6]. For example, Li et al. [28] proposed a grouped deep recurrent residual network to effectively model the complex mapping function from LR HSI to HR HSI. Li et al. [27] introduced a hybrid convolutional model that combines 2-D convolution and separable 3-D convolution to extract spatial and spectral features, respectively. The integration of DL and coherent regularization has opened up new research directions. Building on this, Wang et al. [29] designed a regularization network to enhance spatial–spectral correlation.

B. Spectral Super-Resolution of RGB Image

Spectral super-resolution aims to generate HR HSIs by reconstructing spectral channels from HR RGB images. Traditional algorithms achieve HR HSIs generation by leveraging sparse representations of data. Parmar et al. [8] first used sparse recovery to reconstruct HR HSIs from RGB images. Building on this approach, Arad and Ben-Shahar [30] proposed to compute the dictionary representation of each RGB pixel through an orthogonal matching pursuit algorithm. In the absence of a learnable dictionary, Nguyen et al. [31] put forward a strategy to train a radial basis function network for recovering scene reflectance through spectral transformation. Recently, increasing attention has been directed toward DL methods to enhance super-resolution performance. Significant progress has been made using CNNs to learn the mapping relationship from RGB images to HR HSIs [9], [10], [11], [32], [33], [34]. For instance, Fubara et al. [9] and Zhang et al. [10] used auxiliary spectral bases to provide spectral reference information. Additionally, Hu et al. [32] designed a spatial–spectral attention module with frequency-domain analysis to reduce discrepancies between input and output in the frequency domain. The transformer has been applied to spectral super-resolution to capture inherent self-similarity and long-range dependencies [35], [36]. Cai et al. [36], for example, introduced a spectral multihead self-attention module based on spectral self-similarity and implemented a U-shaped structure to extract contextual information for spectral super-resolution.

C. Fusion-Based Super-Resolution

Fusion-based super-resolution aims to generate HR HSIs by fusing the spatial information of RGB images and the spectral information of LR HSIs. Fusion-based methods primarily include traditional techniques such as component substitution (CS) [37], [38], multiresolution analysis (MRA) [39], [40], Bayesian methods [41], matrix decomposition methods [42], [43], [44], as well as DL-based fusion approaches [12], [13], [17], [18], [19], [45], [46], [47], [48], [49]. For example, Selva et al. [40] used MRA to fuse HSIs and multispectral images, creating high-resolution images through a linear regression-based synthesis of multispectral images. Dian et al. [48] proposed a zero-shot HSI sharpening method that generates reliable training data by accurately estimating the spectral and spatial degradation model of the imaging sensor. Liu et al. [49] proposed a progressive spectral mapping and adaptive loss to improve HSI fusion via multistage fitting and discrepancy-aware refinement. Hu et al. [17] introduced FusFormer, which leverages the global modeling capability of transformers and incorporates residual learning to enhance HSI super-resolution. Jia et al. [18] proposed MSST-Net, a dual-branch multiscale spatial–spectral transformer network designed to effectively integrate spatial and spectral information. Cao et al. [19] proposed an unsupervised fusion-based super-resolution framework, uHNTC, which integrates both transformer and CNN architectures. He et al. [45] modeled the long-range dependence using Mamba to achieve the interaction and fusion of cross-modal information.

All of the above methods are based on the assumption that the images are paired and aligned. To overcome this limitation, recent studies [14], [15], [16], [50], [51], [52], [53], [54] have focused on developing models to handle unaligned data. Zheng et al. [14] developed a nonrigid registration HSSR network, which addresses complex mismatches through random pixel shifts. In contrast to approaches like [50] and [51] that address global rigid geometric deformations, the alignment module in [15] estimates dense optical flow maps to achieve pixel-level transformations for each reference layer, making it more robust to nonrigid deformations. Furthermore, Guo et al. [54] proposed a unified HSI fusion framework that jointly optimizes image registration, blur kernel learning, and image fusion to address variations in satellite imaging conditions. However, these models still require paired HS–RGB images as inputs. To reduce data costs and increase model applicability, Qu et al. [20] utilized spectral unmixing for RGB image spectral reconstruction, where abundances were derived from RGB images and endmembers guided by an HR HSI.

Analyzing the related works in the HSSR task, we draw the following conclusions.

1) Most fusion-based HSSR methods rely on paired HS–RGB datasets, the acquisition of which is both challenging and costly in real-world scenarios.
2) Existing unpaired fusion methods [7], [20], [23] adopt data-driven deep networks for HSI reconstruction, yet neglect the physical significance of spectral reflectance properties.

The differences between our proposed method and existing works lie in two key aspects.

1) We propose a novel unsupervised network designed specifically for the fusion of unpaired HS–RGB images, reducing reliance on pairwise images.

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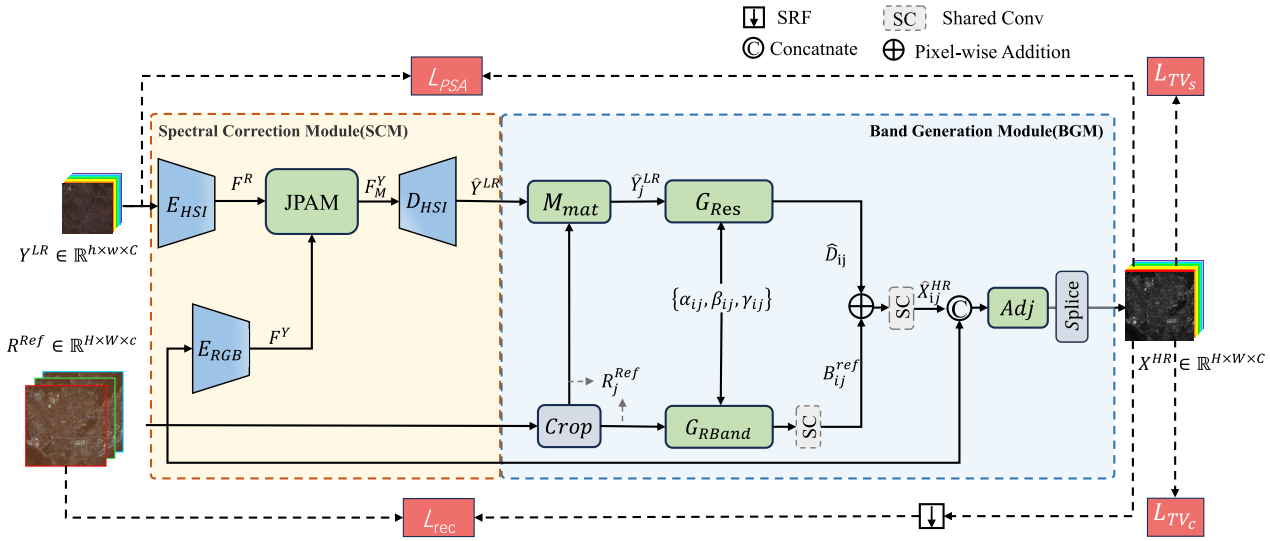


Fig. 2. Overview of our U^{2}SRnet. It takes all RGB images R^{Ref} and a fixed LR HSI Y^{LR} as inputs. The SCM utilizes JPAM to correct the spectral features of LR HSI, ensuring material reflectance consistency. The BGM first selects HS–RGB block pairs from R^{Ref} and $\hat{Y}^{\text{LR}}$, which are most likely to exhibit high material-semantic similarity. Subsequently, the devised differential migration mechanism achieves the differential representation and migration of reflectance properties.

2) The overall design of our framework focuses on modeling spectral reflectance properties, aiming to enhance reconstruction accuracy and improve algorithm generalization.

III. Methodology

We introduce design principles and details of the proposed method in this section. In Section III-A, we define necessary notations and introduce the framework of our proposed method. Next, in Sections III-B and III-C, we delineate each module of the proposed method. The loss function is introduced in Section III-D.

A. Problem Formulation and Overview

Given an HSI with LR $Y^{\text{LR}} \in \mathbb{R}^{h\times w\times C}$ and a reference RGB image with HR $R^{\text{Ref}} \in \mathbb{R}^{H\times W\times c}$, the goal of hyperspectral fusion is to reconstruct the HR HSI $X^{\text{HR}} \in \mathbb{R}^{H\times W\times C}$, where H and h represent the height, W and w represent the width, C and c represent the number of spectral bands, respectively. In general, RGB images have higher spatial resolution, while LR HSIs have higher spectral resolution, i.e., $H > h$, $W > w$, and $C > c$.

An overview of our U^{2}SRnet is illustrated in Fig. 2. The framework includes two modules, which are SCM and BGM, respectively. The SCM is built with four major components: HSI encoder, RGB encoder, JPAM, and HSI decoder. This module encodes the spectral divergence caused by different imaging environments as spectral attention weights to guide the correction of spectral features in the LR HSI, enforcing material reflectance consistency. The BGM is composed of four major components as well: material-matching module (M_{mat}), reference BGM (G_{RBand}), residual generation module (G_{Res}), and adjustment module (Adj). Block pairs with the same materials are identified and selected from unpaired images to facilitate further migration. Subsequently, a differential migration mechanism is employed to migrate the reflectance properties from LR HSI blocks to RGB image blocks.

B. Spectral Correction Module

HSIs contain rich spectral information, which can guide the spectral reconstruction of corresponding materials in RGB images [16], [20], [21]. However, the reflectance properties of the same material may vary across different imaging environments (e.g., light incidence angle and sensor position), as illustrated in Fig. 1(b) and (c). Therefore, our primary task is to leverage the spectral differences between unpaired images to correct the spectral features of the LR HSI for reflectance property consistency. To this end, we design the SCM, which adaptively calibrates the fixed LR HSI spectral features to accommodate varying imaging environments, thereby enhancing the consistency of material reflectance properties. The details of SCM are described as follows.

1) RGB Encoder: We build an RGB encoder with DenseNet as the backbone [55] to effectively extract spectral features while preserving the spatial–spectral feature relationship, as illustrated in Fig. 3(a). The RGB encoder comprises two channel-adjustment blocks (CALs) at the front and end, three dense connection blocks (DCBs), and two transition layers (TLs). The channel adjustment layers preserve the spatial dimensions while adjusting the number of channels. The growth rate of channels in DCBs is set to 10. The first two DCBs contain four convolutional layers, while the last one contains three. The output feature map of the RGB image, denoted as F^R, can be expressed as

$$F^R = E_{\text{RGB}}(R^{\text{Ref}}) \in \mathbb{R}^{m\times n\times l} \tag{1}$$

where $E_{\text{RGB}}(\cdot)$ denotes the RGB encoder, and m, n, and l represent the height, width, and the number of spectral bands of the output feature, respectively. Notably, $l \gg C$.

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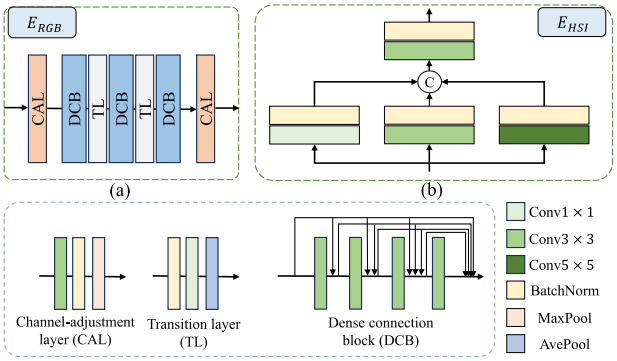


Fig. 3. Illustration of the (a) RGB encoder and (b) HSI encoder.

2) HSI Encoder: The LR HSI possesses high spectral resolution but lacks spatial details. Therefore, we construct a parallel feature extraction network with convolutional kernels of sizes $k = 1, 3, 5$, as shown in Fig. 3(b). Using convolutional kernels of varying sizes effectively captures multiscale texture details, maximizing the extraction capacity of spatial features while preserving the quality of spectral features. The output feature map of the LR HSI, denoted as F^Y, can be expressed as

$$F^Y = E_{\mathrm{HSI}}(Y^{\mathrm{LR}}) \in \mathbb{R}^{m\times n\times l} \tag{2}$$

where $E_{\mathrm{HSI}}(\cdot)$ denotes the HSI encoder.

3) JPAM: This module is designed to enforce material reflectance consistency across the feature dimension, as illustrated in Fig. 4. Given the feature map F^R for the RGB image and F^Y for the LR HSI, we first generate joint feature map $F^D \in \mathbb{R}^{m\times n\times l}$, which captures the spectral divergence caused by different imaging environments, by subtracting two feature maps. Subsequently, F^D is converted into channel attention map $M^D \in \mathbb{R}^{1\times 1\times l}$ through pooling layers followed by shared convolutional layers. By encoding the differences in imaging environments and the resulting spectral divergence as attention weights, M^D, which represents the correction coefficients for the high-dimensional spectral features of LR HSI, is injected into F^Y to enhance the material reflectance consistency, producing the final corrected LR HSI feature map $F_M^Y \in \mathbb{R}^{m\times n\times l}$.

This process can be expressed as

$$F^D = F^R - F^Y \tag{3}$$

$$s_1 = \mathcal{F}_{\mathrm{GAP}}\left(F^D\right) = \frac{1}{m\times n}\sum_{i=1}^{m}\sum_{j=1}^{n} F^D(i,j) \tag{4}$$

$$s_2 = \mathcal{F}_{\mathrm{GMP}}\left(F^D\right) = \max\left(F^D(i,j)\right) \tag{5}$$

$$z_1 = \mathcal{F}_{\mathrm{SC}}(s_1) \tag{6}$$

$$z_2 = \mathcal{F}_{\mathrm{SC}}(s_2) \tag{7}$$

$$M^D = \sigma(z_1 + z_2) \tag{8}$$

$$F_M^Y = F^Y + F_M = F^Y \otimes \left(1 + M^D\right) \tag{9}$$

where $\mathcal{F}_{\mathrm{GAP}}(\cdot)$ and $\mathcal{F}_{\mathrm{GMP}}(\cdot)$ denote the global average pooling and global max pooling, respectively. $s_1, s_2, z_1, z_2 \in \mathbb{R}^{1\times 1\times l}$ are intermediate variables, and σ denotes the sigmoid function.

It is worth noting that JPAM adaptively adjusts M^D based on the joint feature map F^D, enabling effective adaptation to different RGB images and their corresponding imaging environments.

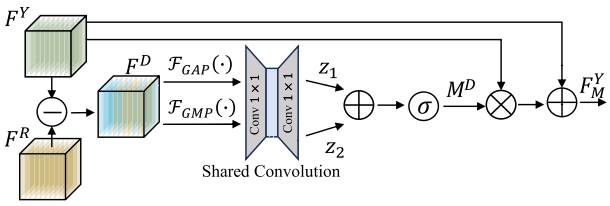


Fig. 4. Illustration of the JPAM.

4) HSI Decoder: We design an HSI decoder to generate the corrected LR HSI from the feature map F_M^Y. This decoder comprises three convolutional layers, followed by a max-pooling layer. The corrected LR HSI is denoted as $\hat{Y}^{\mathrm{LR}} \in \mathbb{R}^{h\times w\times C}$.

C. Band Generation Module

Existing fusion-based methods [12], [13], [14], [15], [16] fail in unpaired scenarios due to spatial misalignment of materials, and physical priors play a critical role in unpaired fusion. Accordingly, we propose the BGM, a hybrid feature-data domain framework that synergistically combines material-semantic matching in deep shared feature space with data-domain differential migration of spectral reflectance properties.

1) Material-Semantic Matching: With RGB images and the corrected LR HSI as inputs, the objective of the material-matching mechanism is to identify, for each RGB block, the block from $\hat{Y}^{\mathrm{LR}}$ with the highest material similarity for the subsequent migration. Specifically, sliding window operation is applied on R^{Ref}, cropping it into δ blocks, each denoted as $R_j^{\mathrm{Ref}} \in \mathbb{R}^{h_1\times w_1\times 3}$, where j denotes the jth block of the data, and h_1 and w_1 are the height and width of the block, respectively. Subsequently, we develop a material-matching module, denoted as M_{mat}, which embeds the inputs into the deep semantic feature space and utilizes the nonnegative Pearson CC to identify and acquire the hyperspectral block $\hat{Y}_j^{\mathrm{LR}}$ with the highest semantic similarity to the RGB block R_j^{Ref}, as illustrated in Fig. 5.

To ensure uniformity in the common feature space, we resize the two inputs to the same size. Specially, we upsample $\hat{Y}^{\mathrm{LR}}$ using bicubic interpolation (BI) method, yielding $\hat{Y}_{\uparrow}^{\mathrm{LR}} \in \mathbb{R}^{h_1\times w_1\times C}$. Notably, the spectral response function (SRF) physically characterizes the spectral mapping between RGB and HS bands. Motivated by the direct utilization of SRF in network design [11], we employ the SRF to generate the spectrally downsampled $\tilde{Y}^{\mathrm{LR}} \in \mathbb{R}^{h_1\times w_1\times 3}$, preserving essential spectral information while matching the size of R_j^{Ref}. We then utilize the pretrained ResNet50 as the shared feature extractor to embed both modalities into a semantically aligned space. By leveraging deep features extracted from a pre-trained ResNet50, our method captures high-level semantic information rather than low-level textures, ensuring that matched regions exhibit consistent material properties even when their raw pixel distributions differ. Specifically, we extract features from R_j^{Ref} and $\tilde{Y}^{\mathrm{LR}}$, obtaining $f_{R_j} \in \mathbb{R}^{(h_1/\rho)\times(w_1/\rho)\times d}$ and $f_{\tilde{Y}} \in \mathbb{R}^{(h_1/\rho)\times(w_1/\rho)\times d}$, respectively, where ρ denotes the size scaling ratio determined by the number of layers used in ResNet50, as detailed in Table I. Next, the feature similarity

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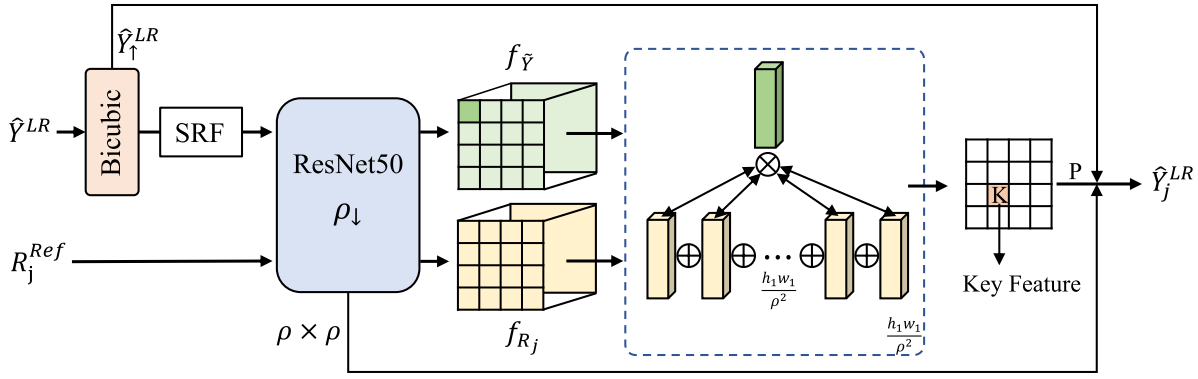


Fig. 5. Illustration of the material-matching module. We utilize ResNet50 as a shared feature extractor to embed the RGB block R_j^{Ref} and the corrected LR HSI $\hat{Y}^{\mathrm{LR}}$ into the common feature space. The nonnegative Pearson CC is then adopted to determine the position of the key feature vector and obtain the hyperspectral block $\hat{Y}_j^{\mathrm{LR}}$.

TABLE I
KEY INFORMATION OF THREE DATASETS

Dataset	Cave	Harvard	TG1HRSSC
Channel number	31	31	54
Wavelength range (nm)	400−700	420−720	400−900
Image size (pixels)	512×512	1040×1392	256×256

is calculated by employing the nonnegative Pearson CC as

$$\begin{aligned} \bar{f}_{R_j} &= f_{R_j} - \mu_{R_j} \\ \bar{f}_{\tilde{Y}} &= f_{\tilde{Y}} - \mu_{\tilde{Y}} \end{aligned} \tag{10}$$

$$c(m,n,k) = \max\left(0, \frac{\left(\bar{f}_{R_j}(m_k,n_k)\right)^T \left(\bar{f}_{\tilde{Y}}(m,n)\right)}{\left\|\bar{f}_{R_j}(m_k,n_k)\right\| \left\|\bar{f}_{\tilde{Y}}(m,n)\right\|}\right) \tag{11}$$

where $0 \le m < (h_1/\rho), 0 \le n < (w_1/\rho)$, and $0 \le k < (h_1 w_1/\rho^2)$. (m_k, n_k) and (m, n) denote the positions of feature vectors in the feature maps f_{R_j} and $f_{\tilde{Y}}$. μ_{R_j} and $\mu_{\tilde{Y}}$ denote the means of the feature maps f_{R_j} and $f_{\tilde{Y}}$.

We calculate the sum of similarities between each feature vector in $f_{\tilde{Y}}$ and all feature vectors in f_{R_j}, choosing the feature vector in $f_{\tilde{Y}}$ with the highest score as the key feature vector, whose spatial position P is identified as follows:

$$P = (m,n) \leftarrow \max\left(\sum_{k=0}^{\frac{h_1 w_1}{\rho^2}-1} c(m,n,k)\right). \tag{12}$$

Benefiting from the fact that convolution operation enables feature maps to preserve spatial contextual relationships, the hyperspectral block $\hat{Y}_j^{\mathrm{LR}} \in \mathbb{R}^{\rho\times\rho\times C}$ can be obtained from $\hat{Y}_{\uparrow}^{\mathrm{LR}}$ based on ρ and position P of the key feature vector.

2) Differential Migration: Given the material-matched block pairs, it is then necessary to migrate reflectance properties from $\hat{Y}_j^{\mathrm{LR}}$ to R_j^{Ref}. Theoretically, inspired by the principle that the differential operation effectively separates informative spectral components while suppressing the influence of absolute magnitude [56], we propose an innovative differential migration mechanism that directly extracts relative spectral variation information in the original data domain and migrates it for subsequent band generation through migration modules.

To be specific, three bands with wavelengths similar to R^{Ref} are selected from $\hat{Y}_j^{\mathrm{LR}}$ to serve as pseudo-reference bands. We then compute weighted residual maps between the remaining bands and these pseudo-reference bands. The acquisition process of residual maps can be expressed as

$$\begin{aligned} D_{ij} = {} & w_{ij1}\left(\hat{Y}_{ij}^{\mathrm{LR}} - \bar{R}_j\right) + w_{ij2}\left(\hat{Y}_{ij}^{\mathrm{LR}} - \bar{G}_j\right) \\ & + w_{ij3}\left(\hat{Y}_{ij}^{\mathrm{LR}} - \bar{B}_j\right) \end{aligned} \tag{13}$$

where D_{ij} denotes the weighted residual map for the ith band of the jth block. $\bar{R}_j, \bar{G}_j$, and $\bar{B}_j$ denote the pseudo-reference bands selected from $\hat{Y}_j^{\mathrm{LR}}$. w_{ij1}, w_{ij2}, and w_{ij3} are the learnable weight parameters.

Let $w_{ij1} + w_{ij2} + w_{ij3} = 1$, $0 \le i \le (C-3)$, the process of generating band $\hat{Y}_{ij}^{\mathrm{LR}}$ based on the weighted residual map and pseudo-reference bands can be formulated as follows:

$$\hat{Y}_{ij}^{\mathrm{LR}} = D_{ij} + w_{ij1}\bar{R}_j + w_{ij2}\bar{G}_j + w_{ij3}\bar{B}_j. \tag{14}$$

To migrate the reflectance intensity variation information, represented as residual maps, to the RGB image, we introduce the RGB block R_j^{Ref} as reference bands. Inspired by (14), we first derive the weighted residual maps and weighted reference maps from $\hat{Y}_j^{\mathrm{LR}}$ and R_j^{Ref}, respectively. Considering the inconsistency of material distribution across block pairs, instead of using simple pixelwise addition, we introduce a shared convolution (SC) layer and a number of shared learnable parameters $\{\alpha_{ij}, \beta_{ij}, \gamma_{ij}\}_{i=0}^{C-3}$, where $0 \le j \le \delta$, and $\alpha_{ij}, \beta_{ij}, \gamma_{ij} \in \mathbb{R}$, to reconstruct the spectral band. Through gradient backpropagation, both shared learnable weights and convolutional layers collaboratively model adaptive correlations between HSI bands and RGB reference bands. The modified version of (14) is given as follows:

$$\hat{X}_{ij}^{\mathrm{HR}} = \mathrm{SC}\left(G_{\mathrm{Res}}\left(\hat{Y}_j^{\mathrm{LR}}\right) + \mathrm{SC}\left(G_{\mathrm{RBand}}\left(R_j, G_j, B_j\right)\right)\right) \tag{15}$$

where $\hat{X}_{ij}^{\mathrm{HR}}$ denotes the coarse reconstructed spectral band. R_j, G_j, and B_j represent the reference bands of R_j^{Ref}. G_{Res} and G_{RBand} are formulated as migration modules. SC denotes the SC layer.

Specifically, G_{Res} consists of two steps: residual acquisition operation and upsampling operation, as illustrated in Fig. 6. First, we compute the residual maps for each band in

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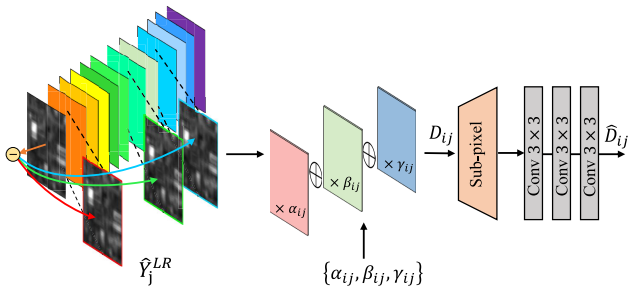


Fig. 6. Illustration of the residual generation module.

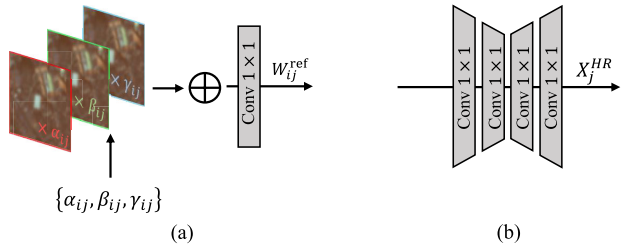


Fig. 7. (a) Illustration of the reference BGM. (b) Illustration of the adjustment module.

$\hat{Y}_j^{\text{LR}}$ relative to the pseudo-reference bands. Then, we introduce $\{\alpha_{ij}, \beta_{ij}, \gamma_{ij}\}$ for weighted computation, yielding $(C-3) \times \delta$ weighted residual maps D_{ij}. The upsampling module comprises a subpixel layer [57], followed by a three-layer convolutional network with a kernel size of 3×3, producing $\hat{D}_{ij}$. In comparison to the transposed convolutional layer, the proposed upsampling module exhibits a larger receptive field, indicating that the output of each $s \times s$ region can be viewed as a pixelwise decomposition of all inputs within the receptive field.

The G_{RBand} module is illustrated in Fig. 7(a). We select R_j^{Ref} as reference bands and introduce shared weight parameters $\{\alpha_{ij}, \beta_{ij}, \gamma_{ij}\}$ along with a 1×1 convolutional layer to obtain the weighted reference map W_{ij}^{ref}. To enhance the consistency of the material distribution between $\hat{D}_{ij}$ and W_{ij}^{ref}, we introduce an SC operation with a 1×1 kernel. First, W_{ij}^{ref} is fed into the SC layer to obtain B_{ij}^{ref}. The sum of B_{ij}^{ref} and $\hat{D}_{ij}$ is then processed through the same SC layer to produce $\hat{X}_{ij}^{\text{HR}}$, which exhibits the same spatial texture as R_j^{Ref}.

Notably, $\hat{X}_{ij}^{\text{HR}}$ is the ith HR hyperspectral band generated using R_j^{Ref} as reference bands. Therefore, we concatenate R_j^{Ref} and each $\hat{X}_{ij}^{\text{HR}}$ following the band sequence in LR HSI, obtaining the full coarse reconstructed HR HSI block $\hat{X}_j^{\text{HR}}$, as expressed in (16). However, since the RGB image does not precisely correspond to the spectral bands at the wavelengths of the final output image, it is essential to design an adjustment module for the spectral adaptation of $\hat{X}_j^{\text{HR}}$, as shown in (17)

$$\hat{X}_j^{\text{HR}} = \text{Cat}\left(\hat{X}_{ij}^0, R_j^{\text{Ref}}\right) \tag{16}$$

$$X_j^{\text{HR}} = \text{Adj}\left(\hat{X}_j^{\text{HR}}\right) \tag{17}$$

where $\text{Cat}(\cdot)$ denotes the concatenation operation, and $\text{Adj}(\cdot)$ indicates the adjustment module.

The Adj module is a lightweight four-layer convolutional network with a kernel size of 1×1, designed to ensure that the spatial texture remains unchanged while adjusting the spectral properties. Convolutional layers are connected through the rectified linear unit (ReLU) activation function. Finally, we splice X_j^{HR} into the complete reconstructed HR HSI X^{HR}.

For a clearer understanding of the workflow, we present the pseudocode of U^{2}SRnet in Algorithm 1.

Algorithm 1 Pseudocode of U^{2}SRnet

Input : RGB image R^{Ref}, fixed LR HSI Y^{LR}
Output: Reconstructed HR HSI X^{HR}

```
for iter ← 1 to epochs do
    Correct Y^LR to generate Ŷ^LR
    Crop the RGB image into δ blocks, denoted as R_j^Ref
    for j ← 1 to δ do
        Select the material-matched block Ŷ_j^LR corresponding to R_j^Ref
        Migrate reflectance properties from Ŷ_j^LR to R_j^Ref, acquiring X̂_j^HR
        Adjust X̂_j^HR to obtain X_j^HR
    end for
    Obtain the final output X^HR by splicing X_j^HR
    return X^HR
end for
```

D. Loss Function

The loss function of U^{2}SRnet is composed of four loss terms, i.e., reconstruction loss L_{rec}, spatial total variation loss L_{TV_s}, spectral total variation loss L_{TV_c}, and pooling spectral angular loss L_{PSA}.

To improve the quality of the reconstruction, we introduce the SRF to spectrally degrade the reconstructed HR HSIs, ensuring that the resulting RGB images are consistent with the original ones. The L_{rec} is defined to minimize the difference between $\mathcal{F}_{\text{SRF}}(X^{\text{HR}})$ and R^{Ref} following L_2-norm as:

$$L_{\text{rec}} = \left\| R^{\text{Ref}} - \mathcal{F}_{\text{SRF}}\left(X^{\text{HR}}\right) \right\|_2 \tag{18}$$

where $\mathcal{F}_{\text{SRF}}(\cdot)$ denotes the spectral degradation using the SRF.

Moreover, we apply total variation loss terms, namely, L_{TV_s} and L_{TV_c}, in both spatial and spectral dimensions separately to suppress noise and ensure spatial smoothness and spectral continuity. The L_{TV_s} is defined by (19), and the L_{TV_c} is defined by (20)

$$L_{\text{TV}_s} = \sum_{m=1}^{C} \sum_{i=1}^{H-1} \sum_{j=1}^{W-1} \Big(\left(X_{i,j+1,m}^{\text{HR}} - X_{i,j,m}^{\text{HR}}\right)^2 + \left(X_{i+1,j,m}^{\text{HR}} - X_{i,j,m}^{\text{HR}}\right)^2 \Big) \tag{19}$$

$$L_{\text{TV}_c} = \sum_{i=1}^{\text{HW}} \sum_{j=1}^{C-1} \left(X_{i,j+1}^{\text{HR}} - X_{i,j}^{\text{HR}}\right)^2. \tag{20}$$

Since our proposed method works in an unsupervised manner, it is impossible to accurately compute the spectral difference between X^{HR} and the ground-truth image. Instead, we constrain the spectral intensity of X^{HR} by minimizing the spectral difference between Y^{LR} and X^{HR}, thereby preventing anomalous reflectance peaks and ensuring the authenticity of

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spectral properties. Specifically, we first apply average pooling to downsample X^{HR}, aligning its spatial dimensions with those of Y^{LR}. We then calculate the cosine similarity of the spectral curves, which can be given by

$$X^{\mathrm{HR}\downarrow} = \mathcal{F}_{\mathrm{AP}}\left(X^{\mathrm{HR}}\right) \tag{21}$$

$$L_{\mathrm{PSA}} = \frac{1}{\hbar w}\sum_{i=1}^{hw}\cos^{-1}\left(\frac{\left(Y_j^{\mathrm{LR}}\right)^T\left(X_j^{\mathrm{HR}\downarrow}\right)}{\left\|Y_j^{\mathrm{LR}}\right\|\left\|X_j^{\mathrm{HR}\downarrow}\right\|}\right) \tag{22}$$

where $\mathcal{F}_{\mathrm{AP}}(\cdot)$ denotes the average pooling.

Our total loss function can be expressed as

$$L = \kappa L_{\mathrm{rec}} + \lambda L_{\mathrm{TV}_s} + \upsilon L_{\mathrm{PSA}} + \xi L_{\mathrm{TV}_c} \tag{23}$$

where κ, λ, υ, and ξ are the tradeoff weights balancing different loss terms.

IV. Experiments

In this section, we perform experiments on three publicly available datasets and discuss the results. First, we present the compared methods, evaluation metrics, datasets, and implementation details in Section IV-A. Then, we conduct the parametric discussion in Section IV-B. Subsequently, we provide a comprehensive performance comparison of U^{2}SRnet with seven other methods in Section IV-C. In addition, complexity analysis and intermediate process visualization are provided in Sections IV-D and IV-E, respectively. Finally, ablation experiments are performed in Section IV-F.

A. Experimental Settings

1) Compared Methods: We compare our proposed method with seven competitors, which are described as follows.

1) *BI:* The BI of spectral dimensions is employed as a baseline method in this work.
2) *UnGUN [20]:* This is a state-of-the-art unsupervised and unpaired approach, which achieves spectral super-resolution via a spectral unmixing strategy. UnGUN takes unpaired RGB images and a fixed HSI as inputs, both of which are with HR.
3) *DoubleSR [7]:* This method employs a semi-supervised and unpaired super-resolution approach to simultaneously enhance the spatial resolution of unpaired low-spatial-resolution RGB and HSIs.
4) *uHNTC [19]:* This unsupervised super-resolution approach integrates both transformer and CNN architectures, incorporating degradation modeling to achieve blind fusion.
5) *M2UNet [58]:* This is an unsupervised method with a three-stage network that blindly fuses LR HSIs and RGB images by incorporating degradation model knowledge and deep image prior.
6) *SDP [23]:* This method introduces an unsupervised spectral diffusion model that guides the inverse diffusion denoising process with spectral distribution priors derived from unpaired HR HSI.
7) *HRSnet [11]:* It is a supervised method that achieves spectral super-resolution by utilizing the SRF as auxiliary guidance.

Among these methods, UnGUN, DoubleSR, and SDP are unsupervised and unpaired approaches. uHNTC and M2UNet are unsupervised and paired methods that train a separate model for each HS–RGB image pair. To ensure a fair comparison, we apply the same training and dataset partitioning strategies across all methods.

2) Evaluation Metrics: Seven quality metrics, i.e., spectral angle mapper (SAM) [59], relative dimensionless global error in synthesis (ERGAS) [60], structural similarity (SSIM) index [61], CC [62], universal image quality index (UIQI) [63], peak signal-to-noise ratio (PSNR) [61], and quality with no reference (QNR) [64] are adopted as the quantitative evaluation metrics.

SAM quantifies spectralsimilarity, ERGAS is an overall fusion quality metric that measures the global mean shifts and dynamic range changes, SSIM captures structural correlation, CC measures the linear dependence between the reference and reconstructed images, UIQI assesses image similarity by considering brightness, contrast, and structural information, and PSNR evaluates the spatial fidelity of the reconstructed image. QNR is a reference-free evaluation metric that measures both spatial and spectral distortions. For SSIM, CC, UIQI, PSNR, and QNR, higher values indicate better quality of the generated image, whereas for SAM and ERGAS, lower values are preferred.

3) Datasets and Implementation Details: We conduct experiments on three public datasets: Cave [65], Harvard [66], and TG1HRSSC [67], with their key information summarized in Table I.

1) *Cave Dataset:* The Cave dataset consists of 32 HSIs, with spectral coverage ranging from 400 to 700 nm. Each image contains 31 spectral bands, sampled at 10-nm intervals, and has a spatial dimension of 512 × 512 pixels.
2) *Harvard Dataset:* The Harvard dataset comprises 50 HSIs, each containing 31 spectral bands and having a spatial dimension of 1040 × 1392 pixels.
3) *TG1HRSSC Dataset:* We use the visible near-infrared HSIs from the TG1HRSSC dataset for experiments. The wavelength ranges from 400 to 900 nm, covering 54 effective bands, and each image has a spatial dimension of 258 × 258 pixels.

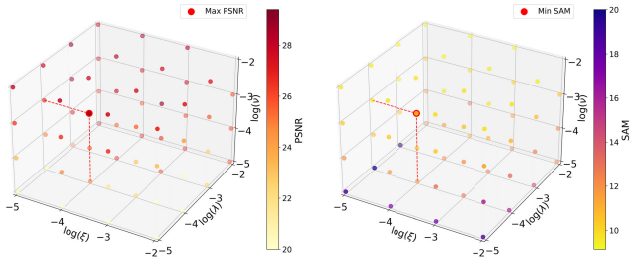
We generate LR HSI and RGB images from the HR HSIs following Wald's et al. [60] protocol. Specifically, we apply a spatial degradation process by performing eightfold downsampling and Gaussian blurring to generate LR HSIs. The RGB images are obtained by applying the SRF to the corresponding HR HSIs. Since the proposed U^{2}SRnet is designed to fuse unpaired HS–RGB images, we randomly select one LR HSI as the fixed hyperspectral input, while utilizing the corresponding RGB images of the remaining HSIs as RGB inputs. For all three datasets, we train the network using a gradient optimization method based on adaptive moment estimation (Adam).

For each dataset, a total of 20 images are randomly selected as the training set, with random flipping applied for data augmentation, while eight images are chosen as the test set. It should be noted that we use the same fixed LR HSI as hyperspectral input for each dataset during both training and testing. The batch size is set to 1, and the total number of training epochs is 600. The initial learning rate is set to 0.003, decaying by a factor of 0.7 every 30 epochs. Notably, during training, the learning rate is adjusted to half of its

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TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THREE DATASETS. THE BEST AND THE SECOND BEST VALUES ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method			Traditional	Non-supervised						Supervised
			BI	uHNTC [19]	M2UNet [58]	DoubleSR [7]	SDP [23]	UnGUN [20]	Ours	HRSnet [11]
Cave	SAM	↓	17.4336	16.2978	15.2757	13.7031	14.9552	11.6164	9.9650	**6.7721**
	ERGAS	↓	14.1096	9.5539	7.1436	6.5841	5.0976	6.6457	3.0472	**2.4284**
	SSIM	↑	0.7230	0.6482	0.8739	0.8201	0.8834	0.8812	0.9111	**0.9524**
	CC	↑	0.8303	0.9223	0.9761	0.9347	0.9806	0.9115	0.9875	**0.9889**
	UIQI	↑	0.6965	0.5581	0.7396	0.7868	0.8564	0.8524	0.8947	**0.9292**
	PSNR	↑	21.3456	23.8277	28.5229	30.0764	30.2755	32.2176	35.7884	**36.5810**
	QNR	↑	0.7043	0.7256	0.7331	0.7517	0.8129	0.8005	**0.8337**	0.8298
Harvard	SAM	↓	10.4192	10.3289	6.2921	7.7453	7.5784	9.9881	6.1124	**3.7973**
	ERGAS	↓	11.2489	6.1208	4.3305	6.5870	3.0936	2.9918	2.5699	**0.9928**
	SSIM	↑	0.7549	0.8601	0.9063	0.8594	0.9407	0.8992	0.9409	**0.9710**
	CC	↑	0.9511	0.9457	0.9548	0.9340	0.9873	0.9535	0.9901	**0.9941**
	UIQI	↑	0.7596	0.8972	0.9352	0.9298	0.9405	0.9411	0.9741	**0.9919**
	PSNR	↑	25.8134	33.6078	34.4662	31.6611	36.9933	36.0914	39.3165	**42.8768**
	QNR	↑	0.7043	0.8034	0.8831	0.8862	0.8694	0.8705	0.9351	**0.9715**
TG1HRSSC	SAM	↓	11.8042	10.6456	7.6018	7.5516	8.1349	7.7032	7.4533	**4.6015**
	ERGAS	↓	7.7724	6.8606	5.5320	2.5842	6.4163	2.0134	2.3195	**1.9983**
	SSIM	↑	0.7881	0.7993	0.8141	0.8015	0.8013	0.8238	0.8408	**0.9028**
	CC	↑	0.7137	0.7946	0.7256	0.7231	0.8205	0.8044	0.8543	**0.8671**
	UIQI	↑	0.8571	0.8587	0.8720	0.9745	0.8466	0.9707	0.9717	**0.9758**
	PSNR	↑	23.8689	24.0111	23.3025	29.0801	25.0311	30.3186	31.1286	**36.2446**
	QNR	↑	0.7936	0.8286	0.8451	0.9338	0.8002	0.9192	0.9216	**0.9535**



(a) (b)

Fig. 8. 3-D visualization of hyperparameter optimization results under different weight combinations of ν, ξ, and λ. (a) PSNR values (higher is better). (b) SAM values (lower is better). The red points denote the optimal configurations.

initial value at the midpoint of the training epochs across all datasets. The tradeoff weights in the loss function are set to $\kappa = 1$, $\lambda = 0.0001$, $\nu = 0.0001$, and $\xi = 0.0001$ on Cave and Harvard datasets, while the $\nu = 0.001$ on the TG1HRSSC dataset, whose selection is described in Section IV-B.

B. Parametric Discussion

The loss function comprises four terms, each targeting a specific optimization aspect of the model. Preliminary experiments indicate that the model exhibits stable performance when the reconstruction loss weight κ is set to 1. Based on this observation, we fix κ to the value of 1 to ensure that the reconstruction loss term maintains its dominant role during optimization.

To optimize the remaining coefficients λ, ν, ξ on the TG1HRSSC dataset, we adopt a grid search strategy over the range $[10^{-5}, 10^{-1}]$, with each step increasing by a factor of 10. We train for 240 epochs and additionally select five images as the validation set to determine the optimal combination of weights. As shown in Fig. 8, the best performance is achieved when λ, ν, and ξ are set to 0.0001, 0.001, and 0.0001, respectively, yielding the highest PSNR and lowest SAM. The same procedure is applied to the other datasets.

C. Results and Comparison

The comparison results used in this article are the optimal outcomes obtained from multiple experiments based on the original articles and source codes.

1) Experimental Results on Cave Dataset: The quantitative results of different methods on the Cave dataset are shown in Table II, with the best results highlighted in bold and the second-best results highlighted in underlined. BI, as the compared baseline, performs the worst among all compared methods. The core of UnGUN, DoubleSR, our U^{2}SRnet, and HRSnet lies in the spectral reconstruction of RGB images. The primary difference is that the first three are nonsupervised paradigms, whereas the latter is a supervised learning strategy. It can be clearly observed that U^{2}SRnet outperforms other nonsupervised methods across all seven evaluation metrics. Specifically, U^{2}SRnet reduces SAM and ERGAS by at least 14.57% and 67.29%, compared to other nonsupervised methods, while PSNR and SSIM improve by at least 11.08% and 3.40%, respectively. HRSnet employs a supervised training strategy, which typically results in higher performance compared to other methods. Our method outperforms HRSnet on the reference-free metric QNR, demonstrating a superior balance between spectral and spatial information.

We utilize SRF to spectrally downsample the super-resolution results to obtain pseudo-color images, as shown in Fig. 9. Due to the limited number of bands in RGB images, BI generates spectral curves that vary monotonically, leading to more pronounced color distortion. As shown in the figure, among the DL-based methods, uHNTC, M2UNet, SDP, and UnGUN produce visually good results. However, the highlighted areas in the error heat map indicate that color distortion

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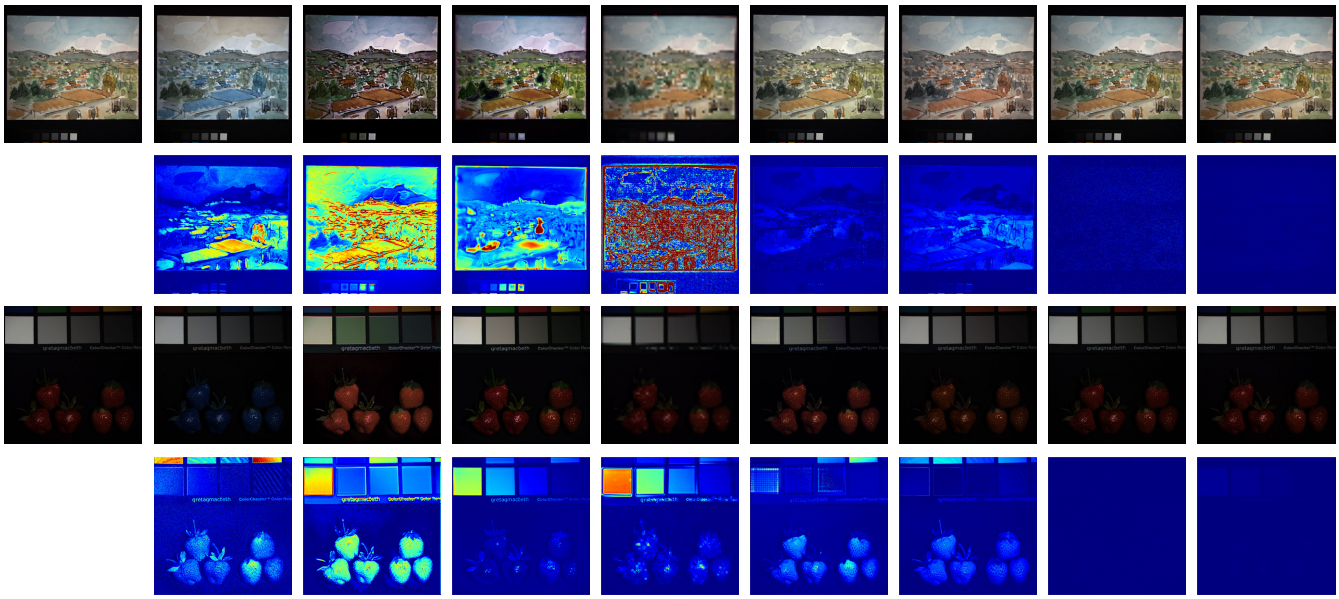


Fig. 9. Visualization results for each compared method on the Cave dataset. (a) Ground truth. (b) BI. (c) uHNTC. (d) M2UNet. (e) DoubleSR. (f) SDP. (g) UnGUN. (h) U^{2}SRnet. (i) HRSnet. Rows I and III display pseudo-color images generated from the ground-truth image and reconstructed images obtained by various methods, while rows II and IV show the error images corresponding to each of the reconstructed images.

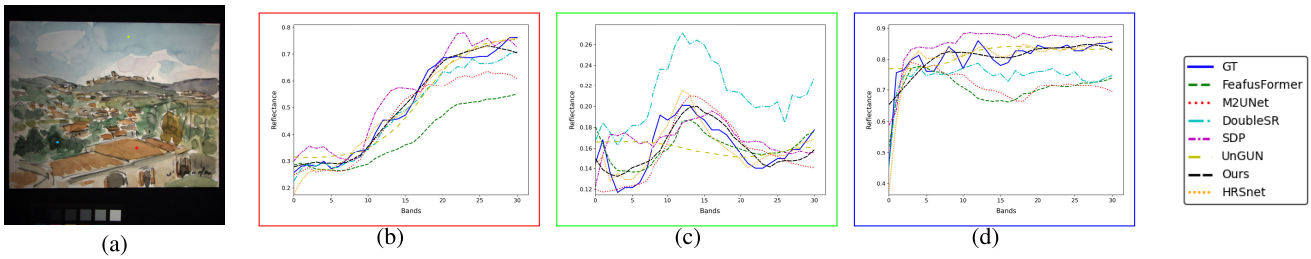


Fig. 10. (a) Pseudo-color image with selected heterogeneous pixels (SAM $>$ 10°) from the Cave dataset. (b)–(d) Spectral curves of different DL-based methods on heterogeneous pixels.

and tonal shifts persist. Additionally, the images generated by the DoubleSR method exhibit poor spatial resolution. The error image of our proposed method is predominantly dark blue, indicating the effectiveness of U^{2}SRnet in super-resolution.

To further assess the performance of each DL-based method, we extract pixel spectra from the reconstructed image and plot spectral curves, as shown in Fig. 10. Since ground-truth material annotations are unavailable for the Cave and Harvard datasets, we define pseudo-material categories based on manually selected visually homogeneous regions (where the average SAM $<$ 10°). In contrast, heterogeneous regions with an average SAM greater than 10° are assigned to different categories. Among the evaluated methods, the curves of U^{2}SRnet are smoother and closer to the reference curves compared to other nonsupervised methods. HRSnet, as a supervised method, has spectral curves that best fit the reference curves.

2) Experimental Results on Harvard Dataset: From the quantitative results (Table II), it is evident that our U^{2}SRnet consistently outperforms the semi-supervised DoubleSR and other unsupervised methods across all seven evaluation metrics. Notably, our method achieves at least 16.42%, 3.51%, 6.28%, and 5.50% performance improvement in ERGAS, UIQI, PSNR, and QNR, respectively. Although there is a gap between U^{2}SRnet and HRSnet, it remains superior among nonsupervised methods.

Fig. 11 presents the visualization results for two random images selected from the Harvard dataset. BI, uHNTC, and UnGUN produce different levels of color distortions. Although UnGUN extracts endmembers from the unpaired guided HR HSI, these endmembers may not be directly applicable to the RGB image due to differences in ground objects. The differences among M2UNet, SDP, U^{2}SRnet, and HRSnet are not significant, as all methods produce visually appealing HR HSIs. The error image reveals that DoubleSR exhibits higher errors in spatially texture-rich regions, which can be attributed to its reduced focus on extracting high-frequency information. The larger image size of the Harvard dataset facilitates the extraction of valid information by our method, leading to superior performance. Although our proposed method still suffers from minor light-coloring errors, its performance is generally comparable to that of HRSnet.

Fig. 12 shows the spectral curves of three heterogeneous pixels selected from the Harvard dataset. As shown in the figure, uHNTC exhibits pronounced spectral distortion, resulting in poorer spectral reconstruction compared to other methods. In contrast, U^{2}SRnet generates spectral curves that

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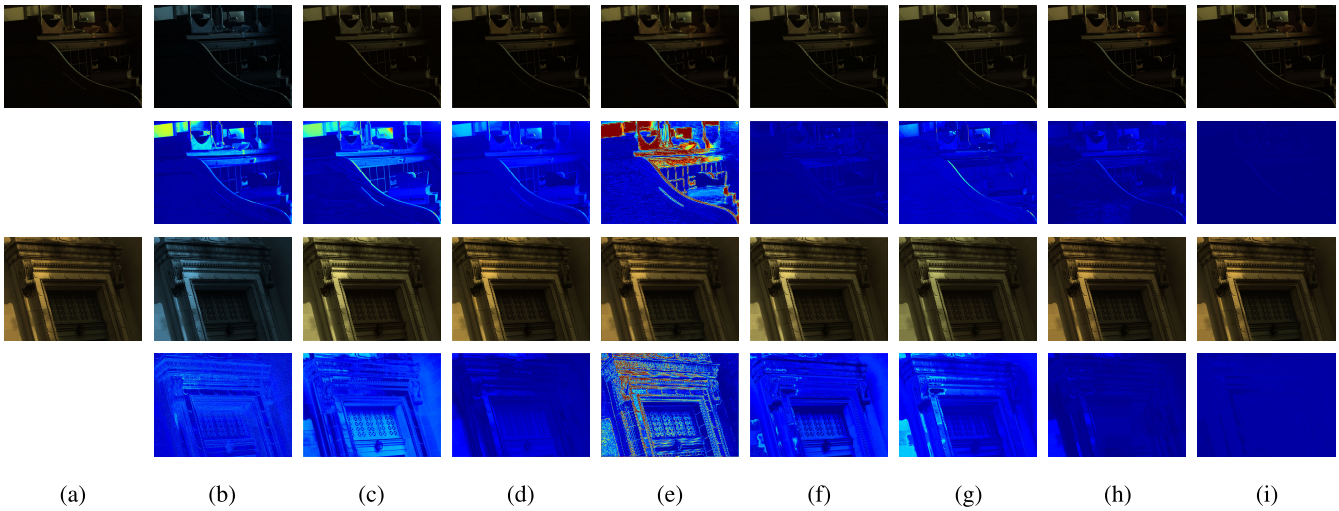


Fig. 11. Visualization results for each compared method on the Harvard dataset. (a) Ground truth. (b) BI. (c) uHNTC. (d) M2UNet. (e) DoubleSR. (f) SDP. (g) UnGUN. (h) U^{2}SRnet. (i) HRSnet. Rows I and III display pseudo-color images generated from the ground-truth image and reconstructed images obtained by various methods, while rows II and IV show the error images corresponding to each of the reconstructed images.

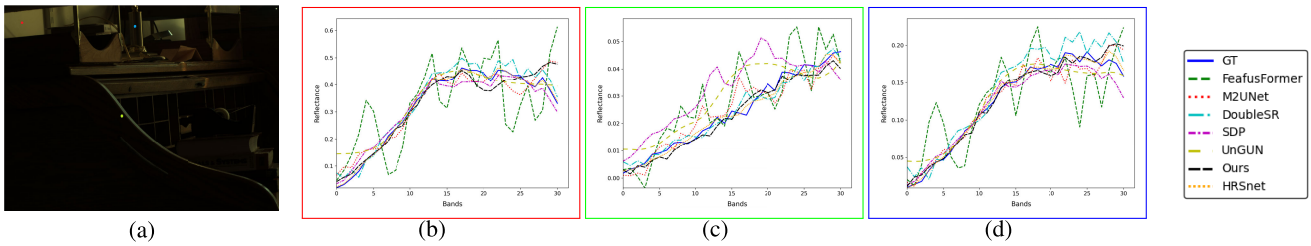


Fig. 12. (a) Pseudo-color image with selected heterogeneous pixels (SAM $>$ 10°) from the Harvard dataset. (b)–(d) Spectral curves of different DL-based methods on heterogeneous pixels.

exhibit a closer alignment with the reference curves, indicating better reconstruction accuracy.

3) Experimental Results on TG1HRSSC Dataset: As shown in Table II, the proposed method achieves competitive performance on the TG1HRSSC dataset, outperforming other nonsupervised approaches in SAM, SSIM, CC, and PSNR, which are critical metrics for assessing spectral fidelity, structural preservation, and pixel-level accuracy. Although slight performance gaps are observed in ERGAS, UIQI, and QNR, likely due to their higher sensitivity to variations in real-world scenarios, the proposed method still demonstrates competitive overall performance.

Fig. 13 presents the visualization results for two random images selected from the TG1HRSSC dataset. From the pseudo-color images, we observe that both our proposed U^{2}SRnet and HRSnet exhibit richer spatial textures than other methods. On randomly selected test images, the error maps produced by our method outperform those of HRSnet.

Fig. 14 presents the spectral curves of three distinct material types in the TG1HRSSC dataset, i.e., building, water, and metal. As shown in the figure, compared DL-based methods achieve satisfactory spectral reconstruction for building materials, which are abundant in the image. However, their ability to recover metal spectra remains limited due to the sparse distribution of this material. In contrast, our method demonstrates robust spectral reconstruction across diverse material types, achieving particularly superior performance compared to other nonsupervised methods.

TABLE III
COMPUTATIONAL COST OF DL-BASED METHODS ON TG1HRSSC DATASET MEASURED BY TOTAL/TRAINABLE PARAMETERS AND FLOPS

Method	uHNTC	M2UNet	DoubleSR	SDP	UnGUN	HRSnet	Ours
Total	20.16M	14.10M	0.42M	4.59M	0.06M	1.50M	9.25M
Trainable	20.16M	14.10M	0.42M	4.59M	0.055M	1.50M	0.98M
FLOPs	119.25G	752.42G	44.39G	1.19 G	4.35G	1.43G	15.03G

D. Complexity Analysis

We analyze the computational complexities of all DL-based methods in this section, where total parameters, trainable parameters, and the number of floating-point operations per second (FLOPs) are employed as the evaluation metrics.

Table III lists the values of three complexity evaluation metrics over the compared approaches. As tabulated in the table, our method shows 0.98M trainable parameters, 9.25M total parameters, and 15.03G FLOPs, ranking mid-range among all compared methods. It is worth noting that we freeze the Conv2, Conv3, and Conv4 layers (Conv2_3_4) in the ResNet50 network, by which gradient computations are eliminated during training, and the number of trainable parameters is significantly reduced. Considering that our proposed method yields the best performance among all nonsupervised methods, it is a decent compromise between performance and computational complexities.

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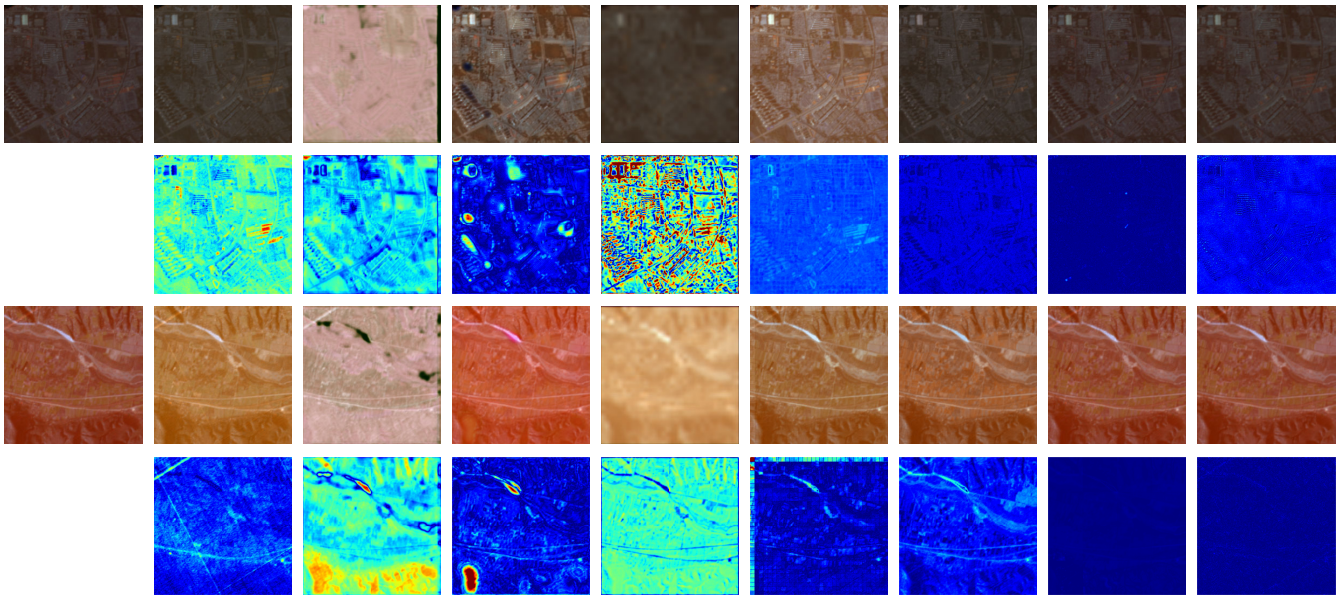


Fig. 13. Visualization results for each compared method on the TG1HRSSC dataset. (a) Ground truth. (b) BI. (c) uHNTC. (d) M2UNet. (e) DoubleSR. (f) SDP. (g) UnGUN. (h) U^{2}SRnet. (i) HRSnet. Rows I and III display pseudo-color images generated from the ground-truth image and reconstructed images obtained by various methods, while rows II and IV show the error images corresponding to each of the reconstructed images.

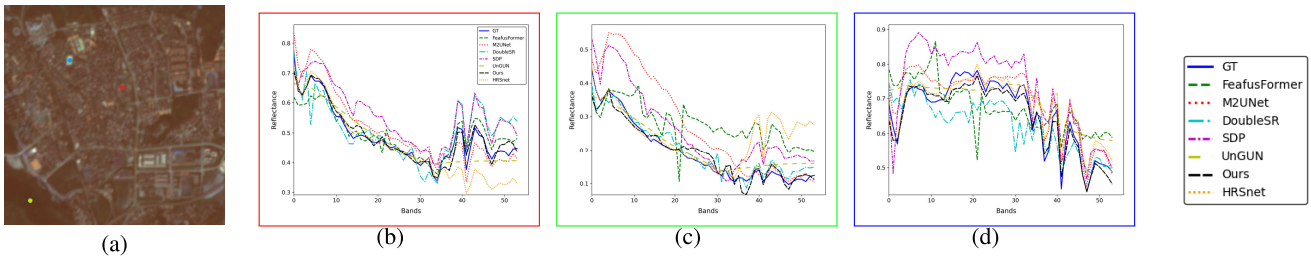


Fig. 14. (a) Pseudo-color image from the TG1HRSSC dataset, with pixels annotated by material type. (b)–(d) Spectral reflectance curves of different materials. (b) Building. (c) Water. (d) Metal.

E. Visualization

Next, to intuitively verify the effectiveness of the material-matching module, we visualize the material-matching results on the TG1HRSSC dataset.

The material-matching results of material-matching module on TG1HRSSC dataset are illustrated in Fig. 15. For intuitive visualization, we display the matching results on pseudo-color images of the HR HSI, which can be degraded to the LR HSI, as shown in Fig. 15(a). Additionally, we scale them for better visualization. For instance, in Fig. 15(c), the RGB block to be matched contains objects as buildings, bridges, and rivers, which correspond to materials metal, cement, and water, respectively. It can be observed the matched hyperspectral block precisely contains the same materials, which substantiates the effectiveness of the material-matching module. The matching results for other RGB blocks further demonstrate the reliability of the material-matching module.

F. Ablation Experiments

In this section, we conduct module ablation experiments to evaluate the contributions of the JPAM, the adjustment module, and the pooling spectral angular loss L_{PSA} across three datasets. We further perform hyperparameter ablation experiments to analyze the impact of the key parameter δ on model performance.

1) Effectiveness of JPAM: This experiment is designed to evaluate the effectiveness of JPAM. To be specific, we replace JPAM with a convolutional block attention module (CBAM) whose input is the concatenated feature of the RGB and HSI encoders. The numerical results on three datasets are tabulated in Table IV. It can be observed from the table that the JPAM-based model constantly outperforms the CBAM-based counterpart on all three datasets, which demonstrates that JPAM can capture spectral variation information effectively, and further enhance the accuracy of spectral reconstruction.

2) Effectiveness of Adj Module: To assess the effectiveness of the Adj module, we remove the module from the whole structure and compare the quantitative results with and without it. As shown in Table IV, when Adj module is removed, the SAM values for the Cave, Harvard, and TG1HRSSC datasets increase by factors of 1.94, 2.56, and 1.43, respectively, compared to the full model. This indicates that reconstructed images generated with the Adj-free model exhibit significant spectral distortion compared to the ground truth, which underscores the necessity of the Adj module.

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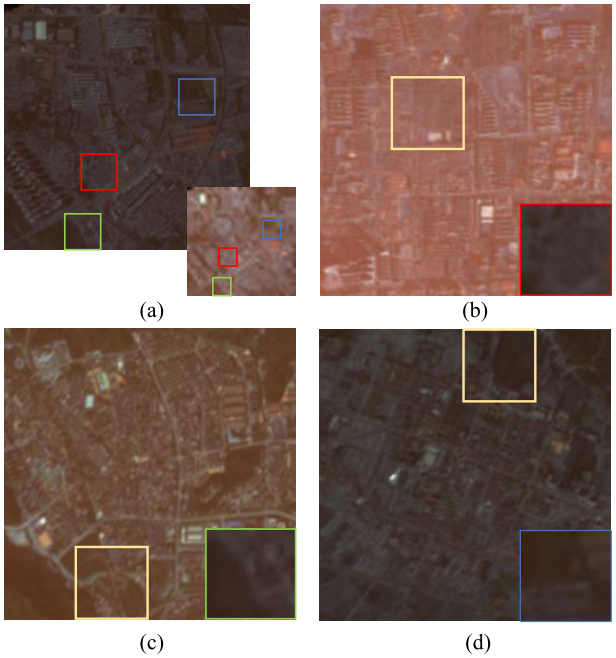


Fig. 15. Visualization results of the material-matching module on TG1HRSSC dataset. (a) Pseudo-color images of the fixed input LR HSI and the HR HSI corresponding to the LR HSI. (b)–(d) RGB blocks and their corresponding hyperspectral blocks. The yellow rectangles represent the RGB blocks to be matched, while the red, blue, and green rectangles correspond to the hyperspectral blocks matched with the RGB blocks.

TABLE IV
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON THREE DATASETS WITH × INDICATING REMOVAL AND **R** INDICATING REPLACEMENT

	JPAM	✓	**R**	✓	✓
	Adj	✓	✓	×	✓
	L_{PSA}	✓	✓	✓	×
Cave	PSNR↑	**35.7884**	33.9635	32.8144	34.4110
	SSIM↑	**0.9111**	0.9053	0.8608	0.8826
	SAM↓	**9.9650**	13.0978	19.2993	20.6301
Harvard	PSNR↑	**39.3165**	38.1920	33.3882	33.7295
	SSIM↑	**0.9409**	0.9231	0.8081	0.8146
	SAM↓	**6.1124**	7.3458	15.6417	16.0332
TG1HRSSC	PSNR↑	**31.1286**	29.9742	28.0745	27.4910
	SSIM↑	**0.8408**	0.8238	0.7501	0.6678
	SAM↓	**7.4533**	7.6707	10.4451	11.9839

3) Effectiveness of L_{PSA}: To demonstrate the effectiveness of the pooling spectral angular loss term, i.e., L_{PSA}, we perform ablation studies by removing this term. The experimental results on all three datasets are presented in Table IV. As shown, the pooling spectral angular loss term enables the U^{2}SRnet to recover a higher quality HR HSI.

4) Impact of Parameter δ: The parameter δ, which denotes the number of blocks in the material-matching module, is an important factor that affects the migration of spectral reflectance properties. To investigate the impact of δ, we conduct experiments on TG1HRSSC dataset by setting δ as five different values, which are 1, 4, 16, 64, and 256, respectively.

The results, presented in Table V and Fig. 16, indicate that the performance of the model improves as δ increases from 1 to 16 but declines with a further increase of δ. The best performance is achieved when δ takes the value of 16. When δ takes a small value, such as 1, the corresponding spatial block size is large. In this scenario, the hyperspectral block fails to provide the complete reflective properties of the desired material, which accounts for the low performance of the model. On the contrary, when δ takes a large value, such as 256, the localized contextual information within a small RGB block is insufficient to encode material semantics, leading to erroneous block-matching results. Therefore, we take $\delta = 16$ as the default setting for all experiments on three datasets.

TABLE V
QUANTITATIVE RESULTS WITH DIFFERENT δ ON THE TG1HRSSC DATASET. BEST VALUES ARE HIGHLIGHTED IN BOLD

δ	SAM↓	ERGAS↓	SSIM↑	CC↑	UIQI↑	PSNR↑
$\delta = 1$	7.6637	2.3942	0.7939	0.8504	0.9701	29.9226
$\delta = 4$	7.5729	2.3868	0.8256	0.7514	0.9695	30.7019
$\delta = 16$	**7.4533**	**2.3195**	**0.8408**	**0.8543**	**0.9717**	**31.1286**
$\delta = 64$	7.6133	2.3653	0.8163	0.8379	0.9703	29.8926
$\delta = 256$	7.7897	2.3831	0.7961	0.8136	0.9685	29.6750

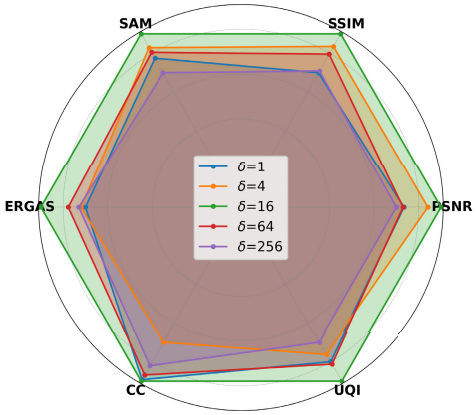


Fig. 16. Performance comparison across different δ on six evaluation metrics.

V. CONCLUSION

In this article, we propose an unsupervised framework, U^{2}SRnet, to tackle the fusion-based super-resolution problem with unpaired HS–RGB images. To enforce material reflectance consistency, we introduce the SCM that corrects the spectral features of LR HSI by leveraging spectral divergence features caused by imaging environment differences between unpaired inputs. Aiming at effectively migrating reflectance properties, we develop the BGM, where HS–RGB block pairs with high material-semantic similarity are acquired, and material properties are migrated between these block pairs in a differential representation mechanism. Extensive experiments on three public datasets demonstrate the superior performance of our proposed method.

In future work, we aim to focus on three directions: 1) we will establish generalizable fusion frameworks for unpaired data, which can be applied to diverse scenarios; 2) we plan to design a reference-free evaluation metric to enhance the performance of unsupervised algorithms; and 3) we will explore scalable network architectures and multiresolution training strategies to equip the model with full-resolution capability and improve its adaptability across varying spatial resolutions.

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Wei Liu received the B.E. degree in information and communication engineering from Xi'an Jiaotong University, Xi'an, China, in 2024, where he is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering.

His research interests include deep learning and remote sensing image understanding and processing.



Haixia Bi (Member, IEEE) received the B.S. and M.S. degrees in computer science and technology from the Ocean University of China, Qingdao, China, in 2003 and 2006, respectively, and the Ph.D. degree in computer science and technology from Xi'an Jiaotong University, Xi'an, China, in 2018.

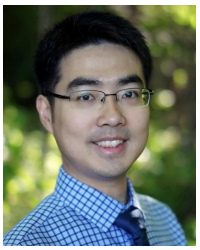
She was a Post-Doctoral Research Fellow with the University of Derby, Derby, U.K., from 2018 to 2019, and the University of Bristol, Bristol, U.K., from 2019 to 2021. She is currently an Associate Professor with the School of Information and Communications Engineering, Xi'an Jiaotong University. Her research interests include machine learning and remote sensing image processing.

Dr. Bi was a recipient of the Best Reviewer Award of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2019 and 2021. She serves as a Guest Editor for several *Remote Sensing* special issues.



Xuehu Zhu received the Ph.D. degree in statistics from Hong Kong Baptist University, Hong Kong, in 2015.

He is currently a Professor with the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an, China. His research interests include statistical learning, high-dimensional data analysis, nonparametric statistics, and change point detection.



Chen Xu received the Ph.D. degree in statistics from the University of British Columbia, Vancouver, BC, Canada, in 2012.

He was a Tenured Faculty Member with the University of Ottawa, Ottawa, ON, Canada, from 2015 to 2023. He is currently a Research Fellow with the Peng Cheng Laboratory, Shenzhen, China. His work has been funded by the Natural Sciences and Engineering Research Council of Canada and the National Key Research and Development Program of China. His research interests include high-dimensional statistics, big data, distributed methods, subsampling, and statistical computing.

Dr. Xu has been serving as an Associate Editor for *Journal of American Statistical Association* and *Electronic Journal of Statistics* since 2023.



Bin Pan (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Astronautics, Beihang University, Beijing, China, in 2013 and 2019, respectively.

He is currently a Professor with the School of Statistics and Data Science, Nankai University, Tianjin, China. He has authored or co-authored over 40 scientific articles. His research interests include cross-domain remote sensing image processing, hyperspectral image classification, unmixing, and super-resolution. Codes for his articles are published online at: https://github.com/Lab-PANbin/

Dr. Pan serves as an Associate Editor for *Infrared Physics and Technology*.

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